

Chapter 6

Hypothesis Testing

Definition 6.2.1 (Best Test)

For testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, with a random sample $X_1, X_2, \dots, X_n$, C_* is called a *best critical region of size α* (or the corresponding test is a *best test of size α*) if:

1. $P(\vec{X} \in C_*; \theta_0) = \alpha$. (This makes it “of size α .”)
2. If C is any other set such that $P(\vec{X} \in C; \theta_0) = \alpha$, then $P(\vec{X} \in C_*; \theta_1) \geq P(\vec{X} \in C; \theta_1)$.

A best test of size α is often referred to as a **most powerful test of size α** .

The Neyman–Pearson Lemma

Suppose that $X_1, X_2, \dots, X_n$ have joint pdf $f(\vec{x}; \theta)$.

Consider the ratio

$$\lambda(\vec{x}; \theta_0, \theta_1) = \frac{f(\vec{x}; \theta_0)}{f(\vec{x}; \theta_1)}.$$

The Neyman–Pearson Lemma (cont)

Let C be the set

$$C = \{\vec{x}: \lambda(\vec{x}; \theta_0, \theta_1) \leq k\},$$

where k is a constant such that

$$P(\vec{X} \in C; H_0) = \alpha.$$

Then C is a *best critical region of size α* for testing

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta = \theta_1.$$

Example 6.2.1

Let $X_1, X_2, \dots, X_n$ be a random sample from the Exponential(θ) distribution. Find the best test of size α for

$H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$,

where $\theta_0 < \theta_1$.

Example 6.2.1 (cont)

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Definition 6.3.1 (UMP)

For simple H_0 and composite H_1 , the critical region C is a **uniformly most powerful critical region of size α** if C is the most powerful (best) critical region for testing H_0 against every simple hypothesis in H_1 .

Theorem 6.3.1 (UMPs and exponential families)

Suppose that the pdf belongs to a one-parameter exponential family:
$$f(\vec{x}; \theta) = a(\theta)b(\vec{x}) \exp[c(\theta)d(\vec{x})].$$

Theorem 6.3.1 (cont)

If $c(\theta)$ is an increasing function of θ , then:

- A UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ is to reject H_0 if $d(\vec{x}) \geq k$,
where k is chosen so that $P(d(\vec{X}) \geq k; \theta_0) = \alpha$.

Theorem 6.3.1 (cont)

If $c(\theta)$ is an increasing function of θ , then:

- A UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta < \theta_0$ is to reject H_0 if $d(\vec{x}) \leq k$,
where k is chosen so that $P(d(\vec{X}) \leq k; \theta_0) = \alpha$.

Theorem 6.3.1 (cont)

If $c(\theta)$ is a decreasing function of θ , then:

- A UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ is to reject H_0 if $d(\vec{x}) \leq k$,
where k is chosen so that $P(d(\vec{X}) \leq k; \theta_0) = \alpha$.

Theorem 6.3.1 (cont)

If $c(\theta)$ is a decreasing function of θ , then:

- A UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta < \theta_0$ is to reject H_0 if $d(\vec{x}) \geq k$,
where k is chosen so that $P(d(\vec{X}) \geq k; \theta_0) = \alpha$.

Definition 6.4.1 (GLR)

Let $L(\theta)$ be a likelihood for a random sample having joint pdf $f(\vec{x}; \theta)$ for $\theta \in \Theta$.

The (generalized) likelihood ratio (GLR) is defined to be

$$\lambda(\vec{x}) = \frac{\max_{\theta \in \Theta_0} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} = \frac{L(\hat{\theta}_0)}{L(\hat{\theta})},$$

where $\hat{\theta}$ denotes the usual “unrestricted MLE” and $\hat{\theta}_0$ denotes the MLE when restricted to the region where H_0 is true.

The corresponding test is a **uniformly most powerful (UMP) test**.

The Generalized Likelihood Ratio Test

Suppose that $X_1, X_2, \dots, X_n$ have joint pdf $f(\vec{x}; \theta)$ with $\theta \in \Theta$. Let $\Theta_0 \subseteq \Theta$.

The generalized likelihood ratio test for

$H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta \setminus \Theta_0$

is to reject H_0 , in favor of H_1 if

$$\lambda(\vec{X}) \leq k.$$

The GLR Test (cont)

- $\lambda(\vec{X})$ is the GLR defined in Definition 6.4.1.

- k is chosen such that

$$\max_{\theta \in \Theta_0} P(\lambda(\vec{X}) \leq k; \theta) = \alpha.$$

Wilks' Theorem

Under certain regularity conditions (given in Section 6.5.1) we have

$$-2 \ln \lambda(\vec{X}) \xrightarrow{d} \chi^2(r - r_0),$$

when H_0 is true.

- r is the number of parameters in the model.
- r_0 is the number of free parameters under the assumption that H_0 is true.

Example 6.5.1

Let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with σ^2 known.

Consider testing

$H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.

What distribution should be use to approximate the test statistic?

Example 6.5.1 (cont)